

Anakin Dey

Curriculum Vitæ

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Research Interests

Algebraic Combinatorics, Algebraic Geometry and Commutative Algebra, Computational Algebra; Algebraic and Fine-Grained Complexity, Algebraic Methods in Combinatorics, Graph Theory, Combinatorial Optimization

Education

2020–2024 **BS in Mathematics**, *University of Illinois Urbana-Champaign*, GPA: 3.94 / 4.00,
Minor in Computer Science
Deans List: Fall 2020, Fall 2021, Spring 2023

Publications

Classifying Primitive Solvable Permutation Groups of Rank 5 and 6

Anakin Dey, Kolton O'Neal, Duc Van Khanh Tran, Camron Upshur, and Yong Yang
2023, *Submitted*, [arXiv:2308.13043](https://arxiv.org/abs/2308.13043)

Post-Disaster Repair Crew Assignment Optimization Using Minimum Latency

Anakin Dey and Melkior Ornik
IEEE International Smart Cities Conference 2022, [arXiv:2206.00597](https://arxiv.org/abs/2206.00597)

Invited Talks

- 2023 **Classifying Primitive Groups using Computer Algebra and Change Making**
UIUC Department of Mathematics Undergraduate Friday Seminars
- 2023 **Classifying Primitive Solvable Permutation Groups of Rank 5 and 6**
China Three Gorges University Algebra and Combinatorics Seminar
- 2022 **Post-Disaster Repair Crew Assignment Optimization Using Minimum Latency**
IEEE International Smart Cities Conference 2022

Research Experience

- Summer 2023 **Undergraduate Research Assistant**, [REU in Algebra, Combinatorics, and Statistics](#),
Under Prof. Yong Yang, Texas State University
- Fall 2021 – **Undergraduate Research Assistant**, [LEADCAT Lab](#), Under Prof. Melkior Ornik,
Present University of Illinois Urbana-Champaign

Student Organizations

Founder and President, ACM [SIGma](#)

Special Interest Group in Math and Algorithms

Admin, ACM [SIGPwny](#)

Special Interest Group in Cybersecurity

Teaching Experience

Spring 2021 – Present **Undergraduate Course Assistant**, University of Illinois Urbana-Champaign

- CS 374 - Introduction to Algorithms and Models of Computation

Spring 2022 - Fall 2022 Course Assistant

Fall 2023 – Present Course Assistant

- CS 173 - Discrete Structures

Spring 2021 - Spring 2022 Course Assistant

Fall 2022 – Present Head Course Assistant

- CS 475 - Models of Computation

Spring 2023 Course Assistant

- CS 125 - Introduction to Computer Science

Spring 2021 Course Assistant

Professional Experience

Summer 2022 **Software Engineering Intern**, CME Group, Chicago, IL

Summer 2021 **Software Engineering Intern**, SiteIQ, Lombard, IL

Selected Coursework

Math 500 Abstract Algebra I

Math 501 Abstract Algebra II

Math 502 Commutative Algebra

Math 511 Introduction to Algebraic Geometry

Math 518 Differentiable Manifolds I

Math 581 Extremal Graph Theory

CS 473 Algorithms

CS 475 Models of Computation

CS 598TMC Algorithms from the Fine-Grained Perspective

CS 598MAF Algebraic and Geometric Complexity Theory

CS 598TH1 Advances in Theoretical CS